Predictive Analysis of Emission-Congestion Decoupling in Intersection-Dense Urban Networks Using Hybrid Data Modeling and XGBoost Learning

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Abstract. Urban transportation networks show complicated relationships between traffic congestion and vehicular emissions, such as CO2 and other gases. Common assumptions suggest that when there is a reduction in congestion, it directly causes a reduction in emission. On the other hand, recent findings indicate the presence of emission-congestion decoupling in certain traffic structures. This work studies the relationship between congestion and emission in intersection-dense areas that addresses the emission-congestion decoupling analysis in intersection dense areas. While usual models assume the linear relationship between congestion and emission, new studies reveal that the relationship is non-linear and dependent. A hybrid analytical predictive model is proposed which integrates real-world traffic travel time data from Washington DC with synthetic grid-based intersection modeling. The model integrates congestion metrics, speed based estimation of emission and machine learning prediction using XGBoost to find out decoupling patterns. The results show that highly congested urban conditions show a strong relationship between emission and congestion where, grid based structured networks show decent decoupling under medium traffic conditions. Feature importance shows that average speed, and intersection delay are the important factors of emission-congestion decoupling.

Keywords: Traffic Congestion, Vehicular Emissions, Emission-Congestion decoupling, XGBoost Classifier.

1 Introduction

The rapid growth of urban areas has intensified problems in transportation in current cities which result in increased congestion and environmental issues. Although traffic management approaches mainly aim to improve mobility, the environmental factors of the reduction of congestion are important and they differentiate based on context. Urban traffic networks with a high density of intersections add difficulties through factors like

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signal timing, queue spillback, and different traffic control methods. Recent studies in mobility analysis have helped in extensive traffic evaluations using probe vehicle data and urban analytics tools. Most current research shows congestion and emission are directly linked, ignoring causes where emission patterns differentiate from congestion trends. Based on this decoupling of emissions and congestion is necessary for sustainable traffic policies. Drawing inspiration form hybrid analytical frameworks used in data pipelines, this study uses a structured methodology which combines urban datasets with synthetic grid modeling. This methodology includes congestion-based assessment, emission estimation, comparative analysis of both, and predictive machine learning. In contrast to available studies on emission predictions and traffic patterns where decoupling takes place. In spite of the advancements in traffic emission modeling, past studies mostly focus on a basic analysis of approaches which include intersection-dense urban environments. Moreover, existing models usually assume a linear relationship between congestion and emission ignoring the factors like speed, delay, volume and intersection that can affect them. This work shows the gap by introducing a hybrid analytical and machine learning model that models emission-congestion decoupling at intersection-dense areas using both real-world and synthetic datasets to improve prediction.

2 Literature Review

2.1 Urban Traffic Emission Modeling Studies

A lot of research efforts have investigated the link between vehicle emissions and traffic behavior using speed-based emission models with traffic analysis frameworks. A lot of studies showed that vehicle emissions respond non-linearly to traffic speeds particularly in stop and go conditions especially in urban areas [1], [2]. Additional research shows these models to analyze how congestion affects fuel consumption and environmental factors in urban areas. These findings shows that average speed and delays are important factors in analyzing emissions and its relationship with congestion [3]. However, most of the past studies have focused more on traffic at a basic level rather than with densely populated urban networks.

2.2 Intersection-Level Traffic and Congestion Research

Intersections create complex traffic patterns like signal delays, queue spillback and various control mechanisms which affect congestion dynamics. Various studies have explored the reason of how both signalized and unsignalized intersections impact the capability of traffic flow and usage of energy [4], [5] . Studies into adaptive signal control and traffic optimization have shown improvements in travelling time but have received mixed results based on emission reductions. Very importantly, at high-volume intersections, continuous engine running has been identified as the factor that impacts emission behaviors [6] [7]. Despite this, there is a need for research which particularly investigates the relationship between emissions and congestions through distinct types intersection structures within an analytical framework.

2.3 Real-World Urban Mobility Data and Emission Analysis

The availability of mobility datasets like floating car data and GPS-based travel time records has helped in detailed urban emission analyses [8], [9]. Previous researches which have used probe vehicle datasets has described the ability of real-world traffic data in understanding urban mobility and their environmental impact. Past studies have integrated travel time analysis with spatial emission mapping to show pollution patterns in densely populated urban areas [10] [11]. Even though these methods show valuable information, many analyses continue to be descriptive [12] and miss predictive tools for identifying traffic patterns which affect emission behavior.

2.4 Machine Learning Approaches in Traffic and Emission Prediction

Machine learning methods have received an increased interest to predict traffic conditions and to estimate emissions. Gradient boosting algorithms like XGBoost have proven useful forecasting congestion levels by using various traffic data characteristics. Researchers have analyzed [13] [14] [15] neural network models to understand temporal traffic forms and their environmental impacts. Anyway, these Methodologies tend to show high levels of predictive accuracy, they mainly focus on calculating emission quantities instead of analyzing decoupling of emissions and congestion behavior [16]. This issue causes the need for predictive frameworks which combine analytical modeling along with machine learning techniques.

2.5 Emission-Congestion Decoupling and Sustainability Studies

Recent studies on sustainability have shown the idea of decoupling, which basically describes scenarios where environmental results differ in economic ad travel patterns. In this context of transportation regimes [17] [18], decoupling refers to situations where when there is a change in congestion, it does not result in a change in emissions, especially where the advancements in traffic flow does not reduce emissions due to factors like acceleration and idling patterns. Most of the research has focused on decoupling at a broader level with intersection-level analyses is not much explored [19] [20]. This study aims to analyze the gap by combining real-world urban data, grid-based modeling of intersections and Machine Learning predictions to analyze decoupling behavior among densely-structures intersections.

3 Research gaps

A review of current research on urban traffic emissions and congestion shows many challenges. Firstly, most recent studies focus more on the corridor-level of traffic assessments without much focus on urban areas where there is a high intensity in intersections where stop and go affect emission factors. Many analytical models depend completely on simulated traffic situations or particularly on real-world data which limits its use in intersection layouts. The lack of hybrid models creates an oversight in the various types of intersections what influence emission and congestion. Current ML models in traffic mainly focus on emission estimation or traffic flow, rather than exploring decoupling patterns of emission and congestion.

4 Problem Statement

Urban traffic management methods usually work under the part where congestion directly leads to a decrease in vehicle emissions. Nonetheless, new sources show that emission reactions are affected by factors like traffic speed, delays at intersections, and the structure of road network. This results in areas where efforts to reduce congestion do not automatically cause lower emissions. The primary challenge is to tackle in this is to create a analytical and predictive model across real and simulated intersection and decoupling using machine learning methods. The goal of the model is to connect descriptive with predictive modeling in urban emissions.

5 Research Objectives

I. *To formulate an Intersection-Centric Emission-Congestion Analysis.*
II. *To create a Hybrid Model using Real and Grid-Based Traffic Networks.*
III. *To create a Predictive Machine Learning Framework for Decoupling Identification.*

6 Methodology

The proposed approach shown in Figure 1 combines analytical modeling along with machine learning to explore the relationship between emission and congestion in urban networks with a high intensity of intersections.

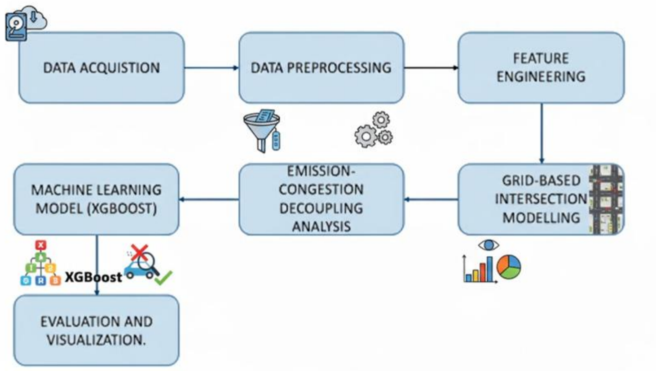

Fig 1. Proposed Workflow.

This approach uses real-world travel time data along with synthetic grid-like traffic simulations to examine the impact of many intersection designs to emission impacts. A comparative analysis of signalized, unsignalized, and high-high traffic intersection environments to identify different structures where emission and congestion are decoupled. This methodology is structured in a way where it includes data preparation, congestion modeling, grid-based simulation, decoupling analysis and predictive learning.

6.1 Real-World Dataset Preparation and Processing

Travel time data from the real world was taken from the Uber Movement dataset which contains origin-destination travel patterns within Washington DC. This dataset offers integrated statistics on travel time, average travel time, lower and upper limits and variations across time across different urban areas [21] [22]. The preprocessing phase involves filtering out certain attributes and standardizing column to allow integration with synthetic datasets. Factors like free-flow travel time and delays are calculated to analyze congestion characteristics. Missing values and inconsistencies are addressed towards feature alignment and normalization, assuring the dataset is compatible with machine learning processes [23] [24]. The final dataset is represented with a baseline for actual urban traffic conditions which is done by dense networks of intersections and congestion.

6.2 Congestion Metric Computation

To analyze the traffic conditions, congestion indicators from observed travel times. Average speed is determined with the relationship between trip length and average time travel, which provides a representation of vehicle [25] movement under varying traffic conditions. Delay is assessed by comparing the actual travel time with a baseline of free-flow conditions which highlights the effects of intersection stops and signal timing on all mobility [26]. Additionally, a congestion index is developed to standardize speed in relation to free-flow conditions, which helps in comparisons across different layouts. These factors serve as the basic analytical sources used for both emission calculation and predictive learning.

6.3 Grid-Based Intersection Modeling

In additional to real-world assessments, synthetic grid structures are developed to duplicate intersection scenarios [27]. The grid formation shows an organized urban structure where origin to destination trips lead to several intersections with varying delay characteristics. Three types of intersections are simulated:

1. *Signalized Intersections:* This represents fixed-time traffic signals which introduces moderate delay.
2. *Unsignalized Intersections*: This represents stop or yield controlled intersections which causes reduced delay and smoother traffic flow.
3. *High-Volume Intersections:* This represents complex multi-intersections with increased congestion

For each setup, travel time and delay metrics are modified to match realistic traffic behaviors. This grid-based modeling technique helps in controlled experimentation to analyze various intersection affect emission-congestion factors. The grid-based model assumes a dew factors where the traffic flows through various intersections. The key assumptions include,

I. *Uniform grid structure representing urban blocks.*
II. *Delay variations depending on intersection types.*
III. *Controlled traffic flow for comparison based analysis.*

6.4 Emission Estimation Approach

Emission factors are generated by using a speed-dependent modeling approach. Based on available literature, vehicle emissions respond non linearly to traffic speed, average speed is used to calculate emission values through a polynomial factor. This method represents increased emissions at low speeds due to stationary running of engine and inefficient acceleration as well as lower emissions during moderate traffic flow. The emission calculation method is applied to both real-world and synthetic datasets for reliability. By connecting emission estimates to congestion metrics, this approach allows for the assessment of emission behavior in various traffic conditions.

6.5 Emission-Congestion Decoupling Analysis

The behavior of decoupling is analyzed by relative shifts in both congestion and emission factors. For every trip data, there is a change is average speed and emission data which is compared with the baseline values to assess whether there is a congestion reduction paired up with emission reduction. Traffic factors are divided into coupled and decoupled systems based on the observed trends. This classification helps in the identification of situations where improvements in traffic flow do not cause environmental changes. This analysis of decoupling proves to give an analytical link between traffic modeling and predictive learning. Emission-Congestion decoupling refers to scenarios where a change in traffic congestion do not result in the proportional changes in vehicular emission. In this study, decoupling is quantified with the decoupling index D1.
We can define the relative shifts for a specific trip i as shown in Equation1:

$$\Delta \text{ Si } = 1 + \frac{\text{Si} - \text{Sbase}}{\text{Sbase}} \; and \; \Delta \text{ Ei} = \frac{\text{Ei} - \text{E Base}}{Ei} \tag{1}$$

where S: Average Speed (Congestion factor) E: Emission Data (Environmental factor) base: The established baseline values.
Decoupling Index (DI), which evaluates the ratio of these changes is represented in Equation 2:

$$\text{DI} = \frac{\Delta \text{ Ei}}{\Delta \text{ Si}} \tag{2}$$

DI > 0 Coupled Emissions and congestion move in the same direction (e.g., faster speeds lead to higher emissions). DI < 0 – Decoupled Improvements in traffic flow (increased speed) occur alongside stable or reduced emissions.

6.6 Machine Learning Prediction Using XGBoost

To increase the analysis above descriptive evaluation, an XGBoost classifier is used to show the decoupling of emissions and congestion through traffic points. The dataset for Machine Learning is created by combining actual real-world data along with grid-based data where it includes characteristics like average speed, congestion index, delay duration, number of intersections and trip distance. Target variables for the decoupling variables are based on analytical modeling results. The XGBoost model is used for its ability to find non-linear patterns in tabular data when it still allows to interpret data through analysis using feature importance. The dataset is split into training and testing portions using stratified sampling to maintain distribution. The model's performance is assessed using classification factors and an analysis of confusion matrix. The results of feature importance analysis is examined to show the main factors causing issues with decoupling prediction, where it connects ideas from Machine Learning with traffic behavior.

6.7 Integration of Analytical Modeling and Predictive Learning

This methodology combines analytical modeling along with supervised machine learning to create a standard framework. Anyways, congestion and emission assessments offer domain knowledge and the XGBoost classification model adds predictive ability which allows the prediction of likelihood of decoupling in new traffic situations. This combined strategy improves the analytical ability of the research by using predictive decision-support model for analyzing traffic at intersections.
The probability of a specific traffic situation x being "decoupled" (y=1) can be expressed as shoen in Equation 3 :

$$\mathrm{P}(\mathrm{y} = 1|\mathrm{x}) \;\; = \sigma \sum\nolimits_{k=1}^{K} fk(\mathrm{Xanalyt} \; + Xraw) \qquad (3)$$

Where: x_{analyt} : The domain knowledge features that is the Decoupling Index DI, relative speed shifts ΔS, and emission factors ΔE derived from the analytical modeling. x_{raw} : The raw traffic data that is volume, time of day, intersection geometry from the Data Acquisition phase. f_k: Represents an individual Decision Tree within the XGBoost ensemble. K: The total number of trees in the model. σ: The Logistic Sigmoid function, which maps the ensemble's output to a probability score between 0 and 1. Predicted Decoupling ŷ = 1 Likely Decoupled and for ŷ = 0 Likely Coupled.

6.8 Hybrid Framework

The proposed model combines analytical models with machine learning to improve prediction. Analytical modeling is used to derive congestion and emission factors, where XGBoost model is trained on both raw and derived features to predict decoupling behavior. This hybrid approach allows the model to integrate domain knowledge along with data driven learning to enable the understanding of complex relationships between emission and congestion Two parameters namely, congestion index and emission estimation are calculated using the Equation 4 and Equation 5, respectively using existing parameters like speed, traffic volume, intersection delay and traffic flow.

$$\boldsymbol{Congestion\ Index} = \mathbf{1} - \frac{\boldsymbol{V}}{\boldsymbol{V_f}} \tag{4}$$

Where,

$V = Actual\ traffic\ speed$
$V_f = Free\ flow\ speed$

$$E = k.\frac{V_{traffic}}{S} \tag{5}$$

Where,
$E = Emission\ level$
$k = Scaling\ constant$
$V_{traffic=} Traffic\ volume$

$S = Speed$

7 Experimental Setup

This setup is designed to analyze emission-congestion relationships among real-world dataset and synthetic dataset. This research uses travel time data from urban transport dataset with grid-based intersection model to estimate traffic regimes that causes emission behavior changes. This mainly focuses on deriving congestion factors, emission factors using speed-based modeling and using a machine learning for predicting emission-congestion decoupling.

7.1 Dataset Details

This research mainly depends on the Uber Movement dataset which shows travel time of vehicles in and around Washington DC. It observes and captures information regarding trip lengths [28] like time travelled, minimum and maximum ranges with shifts around the day by area. Because of this information, researchers can work on the traffic conditions, speeds in normal traffic and delays. Apart from this synthetic grid-based dataset has been created to mirror urban intersections for predictions. Signalized intersections showed stops which caused moderate idling. Unsignalized intersections showed where drivers stopped most often. Each synthetic dataset created for every column created helped in analyzing how long vehicles took to pass through traffic. This setup helped in finding differences based on traffic conditions.

7.2 Environment and Tools

All the scenarios are conducted in a Python environment which is implemented in Google Colab. This completely depended on open-source libraries to assure that it is easily accessible. Data preprocessing and analytical modelling are done using Pandas and NumPy and visualization is performed using Matplotlib and Seaborn for congestion and emission trends. The Machine Learning part is implemented using XGBoost because of its ability to handle tabular data and its capability to give results in non-linear relationships. The dataset is split into training and testing datasets by using stratified sampling to give a balance in class distribution. Model performance is assessed with the help of classification factors and feature importance analyses where it provides interpretability and intersection metrics in emission-congestion decoupling. The workflow in general is executed on a local and cloud workbook environment without the use of high-performance resources. This proves that the proposed approach is computationally efficient. Moreover, and provide dynamic information using multifaceted techniques for Intelligent Transportation System.

8 Results and Discussion

The XGBoost model achieved an accuracy of approximately 99% which indicates high classification performance with minimal miscalculation between coupled and decoupled cases. Feature importance shows the factors like intersection and average speed which affect the emission-congestion decoupling. The analysis of emission-congestion decoupling has resulted in the following outputs which helps solving the case in intersection-dense areas.

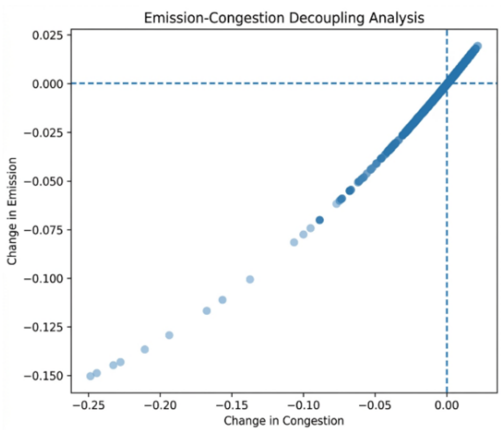

Fig 2. Emission-Congestion Decoupling Behavior for Real-World dataset.

Figure 2 shows the analysis of Washington DC travel time dataset that shows intersection-dense in urban areas that show low average speeds and delay durations. Based on these conditions, the emission factors increase congestion levels that shows a coupled relationship among congestion and vehicular emissions. This helps in the research of stop-and-go traffic conditions that helps in acceleration patterns and increased emissions.

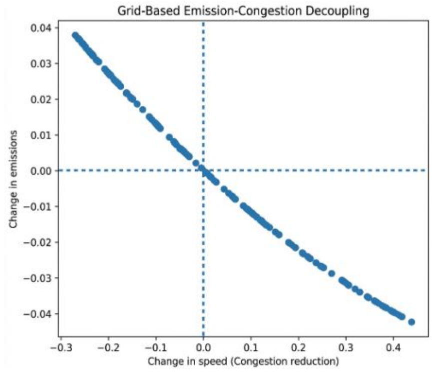

Fig. 3. Grid-Based Emission-Congestions Decoupling for controlled traffic conditions

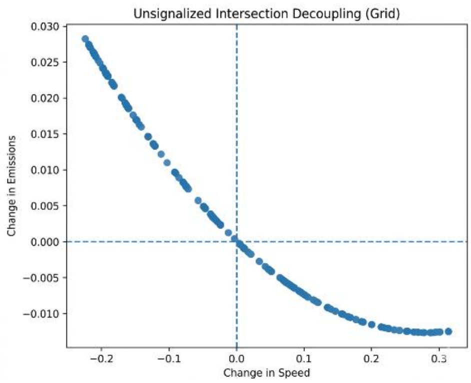

Fig. 4. Grid-Based Emission-Congestions Decoupling for Unsignalized Intersections.

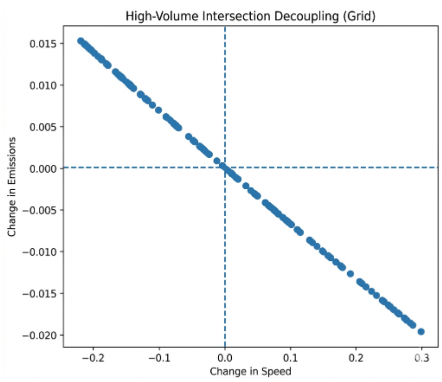

Fig. 5. Grid-Based Emission-Congestions Decoupling for High-Volume Intersections.

Figure 3, Figure 4 and Figure 5 shows grid-based analysis that shows emission-congestion decoupling that shows to be more likely to show with speed systems other than intense congestion or free flow conditions. It also shows the changes in emission-congestion decoupling in Signalized, Unsignalized and High-Volume Intersections. This shows the importance of intersection and traffic control strategies in urban emission behavior.

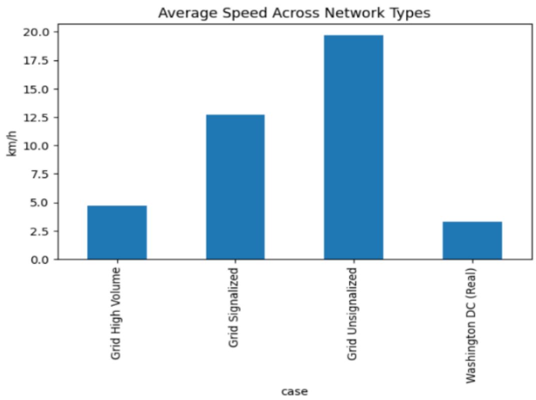

Fig. 6 Average Speed across different networks

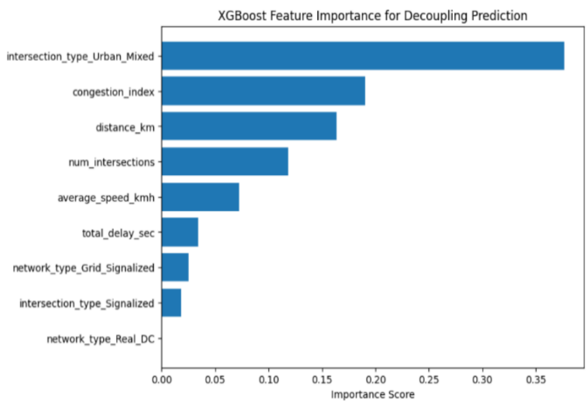

Fig. 7. XGBoost feature importance for Decoupling methods for Traffic dense intersections.

Figure 6 and Figure 7 shows the integrated analysis of real-world and grid-based datasets that shows the non-uniformity of decoupling across different traffic conditions. In heavily congested traffic conditions emissions are increasingly coupled with congestion due to idling and different acceleration patterns. This also shows the non-linear relationship between emission and congestion factors.

9 Conclusion and future scope

This research shows a synthetic analytical and predictive model for estimating emission and congestion relationship in intersection-dense urban traffic systems. Real-world and travel time data from Washington DC is combined with gid-search modeling t ensure that emission behavior is impacted by traffic speed systems and intersection delay factors. The results show that a increasingly congested urban area show coupled emission factors where moderate traffic conditions with grid structures allow the cause of emission-congestion decoupling. The addition of XGBoost model helps in descriptive analysis of the dataset. Future work may include extending this framework through multi-city datasets, real-time traffic aggregation and advanced spatial learning to improve decision-making for urban traffic systems. This work relies on estimated emission values that is derived from traffic parameters other than direct real-time emission measurements, which can affect the accuracy. Future work can focus on the following,

a. *Integration of real-time traffic and emission sensor data.*
b. *Extending the framework to multiple cities.*
c. *Incorporating deep learning techniques to deploy it in smart city management.*

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